```
USE retail_events_db;

SELECT event_id, store_id, product_code, base_price, promo_type
FROM fact_events
WHERE base_price > 1000;
```

Output | Result #2

event_id	store_id	product_code	base_price	promo_type
a1503f	STCBE-1	P15	3000	500 Cashback
6b2afc	STCBE-4	P08	1190	BOGOF
8f25a6	STHYD-6	P15	3000	500 Cashback
0f422c	STBLR-0	P14	1020	BOGOF
7fc923	STTRV-0	P08	1190	BOGOF
ca7298	STCBE-2	P15	3000	500 Cashback
bf33ae	STMYS-0	P14	1020	BOGOF
4f0587	STMDU-0	P08	1190	BOGOF
fa5b45	STVJD-1	P08	1190	BOGOF
7e9777	STVJD-1	P08	1190	BOGOF

OK, 300 records retrieved in 1.488ms

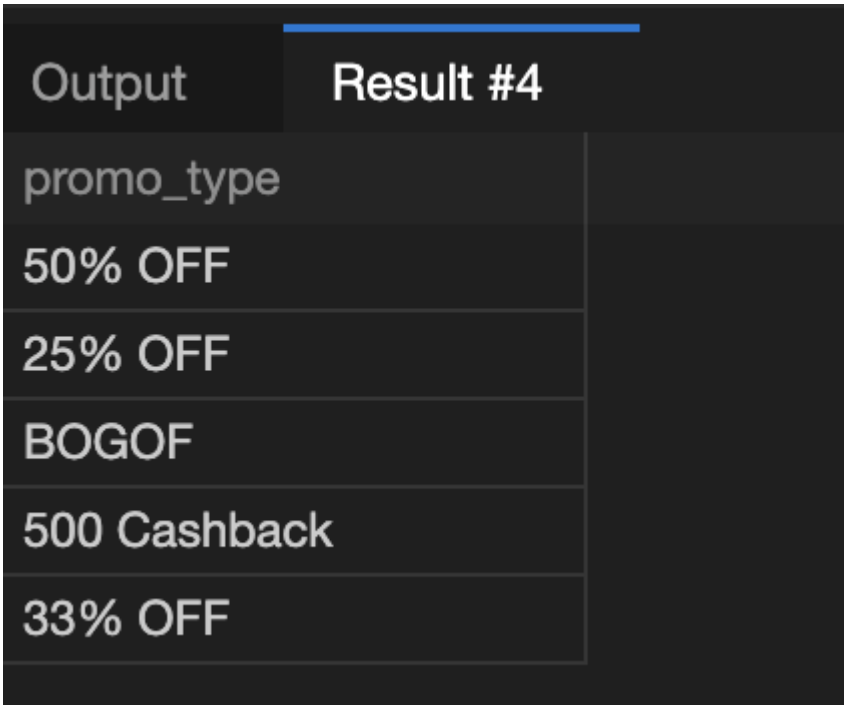
```
-- Q2
SELECT event_id, product_code, promo_type,
      `quantity_sold(before_promo)`, `quantity_sold(after_promo)`
FROM fact_events
WHERE `quantity_sold(after_promo)` > 100
ORDER BY `quantity_sold(after_promo)` DESC;
```

Output | Result #3

event_id	product_code	promo_type	quantity_sold(before_promo)	quantity_sold(after_promo)
e93b62	P04	BOGOF	513	2067
1a1a02	P04	BOGOF	465	2064
fe94ae	P04	BOGOF	450	1984
4eb35e	P04	BOGOF	483	1907
fe7b5d	P03	BOGOF	472	1902
0efde2	P03	BOGOF	448	1895
07e4ea	P04	BOGOF	486	1890
855f49	P04	BOGOF	481	1890
0c9f61	P03	BOGOF	433	1883
6d65d2	P04	BOGOF	480	1867

OK, 979 records retrieved in 1.802ms

```
-- Q3
SELECT DISTINCT promo_type
FROM fact_events;
```




```
-- Q4
SELECT COUNT(*)                      AS total_events,
       SUM(`quantity_sold(before_promo)`) AS total_before,
       SUM(`quantity_sold(after_promo)`)  AS total_after,
       AVG(base_price)               AS avg_price,
       MAX(base_price)               AS max_price,
       MIN(base_price)               AS min_price
FROM fact_events;
```




```
-- Q5
SELECT promo_type,
       COUNT(*)                      AS event_count,
       SUM(`quantity_sold(before_promo)`) AS total_before,
       SUM(`quantity_sold(after_promo)`)  AS total_after
FROM fact_events
GROUP BY promo_type
ORDER BY total_after DESC;
```

Output | Result #6

promo_type	event_count	total_before	total_after
BOGOF	500	58180	215253
33% OFF	200	63321	90576
500 Cashback	100	22299	63180
25% OFF	400	44007	38290
50% OFF	300	21243	28174

```
-- Q6
SELECT promo_type,
	SUM(`quantity_sold(before_promo)`) AS total_before,
	SUM(`quantity_sold(after_promo)`)  AS total_after,
	SUM(`quantity_sold(after_promo)`) - SUM(`quantity_sold(before_promo)`) AS
quantity_change
FROM fact_events
GROUP BY promo_type
ORDER BY quantity_change DESC;
```

Output | Result #6

promo_type	total_before	total_after	quantity_change
BOGOF	58180	215253	157073
500 Cashback	22299	63180	40881
33% OFF	63321	90576	27255
50% OFF	21243	28174	6931
25% OFF	44007	38290	-5717

```
-- Q7
SELECT product_code, product_name, category,
	SUM(`quantity_sold(after_promo)`) AS total_after
FROM fact_events f
JOIN dim_products p USING (product_code)
GROUP BY product_code, product_name, category
ORDER BY total_after DESC;
```

Output | Result #7

product_code	product_name	category	total_after
P04	Atliq_Farm_Chakki_Atta (1KG)	Grocery & Staples	81290
P03	Atliq_Suflower_Oil (1L)	Grocery & Staples	74478
P15	Atliq_Home_Essential_8_Product_Combo	Combo1	63180
P02	Atliq_Sonamasuri_Rice (10KG)	Grocery & Staples	53235
P01	Atliq_Masoor_Dal (1KG)	Grocery & Staples	37341
P13	Atliq_High_Glo_15W_LED_Bulb	Home Appliances	29928
P14	Atliq_waterproof_Immersion_Rod	Home Appliances	23685
P07	Atliq_Curtains	Home Care	16317
P08	Atliq_Double_Bedsheet_set	Home Care	15058
P12	Atliq_Lime_Cool_Bathing_Bar (125GM)	Personal Care	10280

OK, 15 records retrieved in 12.083ms

```
-- Q8
SELECT category,
	COUNT(*)			AS event_count,
	SUM(`quantity_sold(before_promo)`) AS total_before,
	SUM(`quantity_sold(after_promo)`)  AS total_after,
	SUM(`quantity_sold(after_promo)`) - SUM(`quantity_sold(before_promo)`) AS
quantity_change
FROM fact_events f
JOIN dim_products p USING (product_code)
GROUP BY category
ORDER BY total_after DESC;
```

Output | Result #8 | Result #9

category	event_count	total_before	total_after	quantity_change
Grocery & Staples	400	126970	246344	119374
Combo1	100	22299	63180	40881
Home Appliances	200	14713	53613	38900
Home Care	400	19764	40832	21068
Personal Care	400	25304	31504	6200

```
-- Q9
SELECT city,
	COUNT(*)			AS event_count,
	SUM(`quantity_sold(before_promo)`) AS total_before,
```

```
    SUM(`quantity_sold(after_promo)`)  AS total_after
FROM fact_events f
JOIN dim_stores s USING (store_id)
GROUP BY city
ORDER BY total_after DESC;
```

Output | Result #9

city	event_count	total_before	total_after
Bengaluru	300	49171	105141
Chennai	240	39505	83273
Hyderabad	210	34363	69399
Coimbatore	150	18150	38900
Mysuru	120	18569	37470
Visakhapatnam	150	17175	33916
Madurai	120	14458	31169
Mangalore	90	7529	14929
Vijayawada	60	5297	11106
Trivandrum	60	4833	10170

OK, 10 records retrieved in 6.112ms

```
-- Q10
SELECT campaign_name, start_date, end_date,
       COUNT(*)                    AS event_count,
       SUM(`quantity_sold(before_promo)`) AS total_before,
       SUM(`quantity_sold(after_promo)`)  AS total_after
FROM fact_events f
JOIN dim_campaigns c USING (campaign_id)
GROUP BY campaign_name, start_date, end_date
ORDER BY total_after DESC;
```

Output | Result #10 | Result #11

campaign_name	start_date	end_date	event_count	total_before	total_after
Sankranti	01/10/2024	01/16/2024	750	98731	252069
Diwali	11/12/2023	11/18/2023	750	110319	183404

```
-- Q11
SELECT category,
       SUM(`quantity_sold(after_promo)`) AS total_after,
       AVG(base_price)                   AS avg_price
FROM fact_events f
JOIN dim_products p USING (product_code)
GROUP BY category
HAVING SUM(`quantity_sold(after_promo)`) > 1000
ORDER BY total_after DESC;
```

Output | Result #11 | Result #12

category	total_after	avg_price
Grocery & Staples	246344	385.0000
Combo1	63180	3000.0000
Home Appliances	53613	685.0000
Home Care	40832	490.0000
Personal Care	31504	102.3750

```
-- Q12
SELECT city, category,
       SUM(`quantity_sold(after_promo)`) AS total_after
FROM fact_events f
JOIN dim_stores   s USING (store_id)
JOIN dim_products p USING (product_code)
GROUP BY city, category
ORDER BY city, total_after DESC;
```

Output | Result #12 | Result #13

city	category	total_after
Bengaluru	Grocery & Staples	59431
Bengaluru	Combo1	15250
Bengaluru	Home Appliances	12909
Bengaluru	Home Care	10006
Bengaluru	Personal Care	7545
Chennai	Grocery & Staples	46639
Chennai	Combo1	12081
Chennai	Home Appliances	10686
Chennai	Home Care	7974
Chennai	Personal Care	5893

OK, 50 records retrieved in 6.727ms

```
-- Q13
SELECT product_name, category,
       SUM(`quantity_sold(before_promo)`) AS total_before,
       SUM(`quantity_sold(after_promo)`)  AS total_after,
       SUM(`quantity_sold(after_promo)`) - SUM(`quantity_sold(before_promo)`) AS
quantity_change,
       ROUND((SUM(`quantity_sold(after_promo)`) - SUM(`quantity_sold(before_promo)`))
            / NULLIF(SUM(`quantity_sold(before_promo)`), 0) * 100, 2) AS pct_change
FROM fact_events f
JOIN dim_products p USING (product_code)
GROUP BY product_code, product_name, category
ORDER BY pct_change DESC;
```

Output | Result #13

product_name	category	total_before	total_after	quantity_change	pct_change
Atliq_waterproof_Immersion_Rod	Home Appliances	6468	23685	17217	266.19
Atliq_High_Glo_15W_LED_Bulb	Home Appliances	8245	29928	21683	262.98
Atliq_Double_Bedsheet_set	Home Care	4203	15058	10855	258.27
Atliq_Curtains	Home Care	4592	16317	11725	255.34
Atliq_Home_Essential_8_Product_Combo	Combo1	22299	63180	40881	183.33
Atliq_Farm_Chakki_Atta (1KG)	Grocery & Staples	32340	81290	48950	151.36
Atliq_Suflower_Oil (1L)	Grocery & Staples	31309	74478	43169	137.88
Atliq_Masoor_Dal (1KG)	Grocery & Staples	26040	37341	11301	43.40
Atliq_Sonamasuri_Rice (10KG)	Grocery & Staples	37281	53235	15954	42.79
Atliq_Doodh_Kesar_Body_Lotion (200ML)	Personal Care	5257	7022	1765	33.57

OK, 15 records retrieved in 8.924ms

```
-- Q14
SELECT campaign_name, promo_type,
       COUNT(*)                        AS event_count,
       SUM(`quantity_sold(before_promo)`) AS total_before,
       SUM(`quantity_sold(after_promo)`)  AS total_after,
       SUM(`quantity_sold(after_promo)`) - SUM(`quantity_sold(before_promo)`) AS
quantity_change
FROM fact_events f
JOIN dim_campaigns c USING (campaign_id)
GROUP BY campaign_name, promo_type
ORDER BY campaign_name, quantity_change DESC;
```

Output | Result #14 | Result #15

campaign_name	promo_type	event_count	total_before	total_after	quantity_change
Diwali	500 Cashback	50	16791	50769	33978
Diwali	BOGOF	200	10024	34461	24437
Diwali	33% OFF	100	29152	43117	13965
Diwali	50% OFF	200	16843	22074	5231
Diwali	25% OFF	200	37509	32983	-4526
Sankranti	BOGOF	300	48156	180792	132636
Sankranti	33% OFF	100	34169	47459	13290
Sankranti	500 Cashback	50	5508	12411	6903
Sankranti	50% OFF	100	4400	6100	1700
Sankranti	25% OFF	200	6498	5307	-1191

OK, 10 records retrieved in 6.479ms

```
-- Q15
SELECT product_name, category,
       SUM(base_price * `quantity_sold(before_promo)`) AS revenue_before,
       SUM(base_price * `quantity_sold(after_promo)`)  AS revenue_after,
       SUM(base_price * `quantity_sold(after_promo)`)
         - SUM(base_price * `quantity_sold(before_promo)`) AS revenue_difference
FROM fact_events f
JOIN dim_products p USING (product_code)
GROUP BY product_code, product_name, category
```

ORDER BY revenue_difference DESC;

Output | Result #15 | Result #16

product_name	category	revenue_before	revenue_after	revenue_difference
Atliq_Home_Essential_8_Product_Combo	Combo1	66897000	189540000	122643000
Atliq_Farm_Chakki_Atta (1KG)	Grocery & Staples	10851800	29100500	18248700
Atliq_waterproof_Immersion_Rod	Home Appliances	6597360	24158700	17561340
Atliq_Sonamasuri_Rice (10KG)	Grocery & Staples	32061660	45782100	13720440
Atliq_Double_Bedsheet_set	Home Care	5001570	17919020	12917450
Atliq_Suflower_Oil (1L)	Grocery & Staples	5599512	14310708	8711196
Atliq_High_Glo_15W_LED_Bulb	Home Appliances	2885750	10474800	7589050
Atliq_Curtains	Home Care	1377600	4895100	3517500
Atliq_Masoor_Dal (1KG)	Grocery & Staples	4478880	6422652	1943772
Atliq_Doodh_Kesar_Body_Lotion (200ML)	Personal Care	998830	1334180	335350

OK, 15 records retrieved in 7.419ms

```
-- Q16
WITH t AS (
  SELECT promo_type,
        SUM(`quantity_sold(before_promo)`) AS total_before,
        SUM(`quantity_sold(after_promo)`)  AS total_after
  FROM fact_events
  GROUP BY promo_type
), pct AS (
  SELECT *, ROUND((total_after - total_before) / NULLIF(total_before, 0) * 100, 2) AS
percentage_change
  FROM t
)
SELECT *,
      CASE WHEN percentage_change >= 50 THEN 'High Impact'
           WHEN percentage_change >= 20 THEN 'Medium Impact'
           ELSE 'Low Impact' END AS performance_category
FROM pct
ORDER BY percentage_change DESC;
```

Output | Result #16 | Result #17

promo_type	total_before	total_after	percentage_change	performance_category
BOGOF	58180	215253	269.98	High Impact
500 Cashback	22299	63180	183.33	High Impact
33% OFF	63321	90576	43.04	Medium Impact
50% OFF	21243	28174	32.63	Medium Impact
25% OFF	44007	38290	-12.99	Low Impact

OK, 5 records retrieved in 4.401ms

```
-- Q17
WITH t AS (
  SELECT category, product_name,
         SUM(`quantity_sold(after_promo)`) AS total_quantity_after
  FROM fact_events f
  JOIN dim_products p USING (product_code)
  GROUP BY category, product_code, product_name
), r AS (
  SELECT *, DENSE_RANK() OVER (PARTITION BY category ORDER BY
total_quantity_after DESC) AS category_rank
  FROM t
)
SELECT * FROM r WHERE category_rank <= 2;
```

Output | Result #17

category	product_name	total_quantity_after	category_rank
Combo1	Atliq_Home_Essential_8_Product_Combo	63180	1
Grocery & Staples	Atliq_Farm_Chakki_Atta (1KG)	81290	1
Grocery & Staples	Atliq_Suflower_Oil (1L)	74478	2
Home Appliances	Atliq_High_Glo_15W_LED_Bulb	29928	1
Home Appliances	Atliq_waterproof_Immersion_Rod	23685	2
Home Care	Atliq_Curtains	16317	1
Home Care	Atliq_Double_Bedsheet_set	15058	2
Personal Care	Atliq_Lime_Cool_Bathing_Bar (125GM)	10280	1
Personal Care	Atliq_Cream_Beauty_Bathing_Soap (125GM)	7697	2

OK, 9 records retrieved in 7.308ms

```
-- Q18
WITH t AS (
  SELECT city, store_id,
         SUM(`quantity_sold(after_promo)`) AS total_quantity_after
  FROM fact_events f
  JOIN dim_stores s USING (store_id)
  GROUP BY city, store_id
), r AS (
  SELECT *, DENSE_RANK() OVER (PARTITION BY city ORDER BY total_quantity_after
DESC) AS city_rank
  FROM t
)
SELECT * FROM r WHERE city_rank <= 2;
```

Output | Result #18 | Result #19

city	store_id	total_quantity_after	city_rank
Bengaluru	STBLR-7	11866	1
Bengaluru	STBLR-6	11601	2
Chennai	STCHE-4	11547	1
Chennai	STCHE-7	11546	2
Coimbatore	STCBE-0	8657	1
Coimbatore	STCBE-2	8535	2
Hyderabad	STHYD-2	10822	1
Hyderabad	STHYD-0	10522	2
Madurai	STMDU-0	8310	1
Madurai	STMDU-2	7685	2

OK, 20 records retrieved in 7.04ms

```
-- Q19
WITH t AS (
  SELECT campaign_name, product_name,
         SUM(`quantity_sold(before_promo)`) AS total_before,
         SUM(`quantity_sold(after_promo)`)  AS total_after,
         SUM(`quantity_sold(after_promo)`) - SUM(`quantity_sold(before_promo)`) AS
quantity_change,
         ROUND((SUM(`quantity_sold(after_promo)`) - SUM(`quantity_sold(before_promo)`))
              / NULLIF(SUM(`quantity_sold(before_promo)`), 0) * 100, 2) AS
percentage_change
  FROM fact_events f
  JOIN dim_campaigns c USING (campaign_id)
  JOIN dim_products  p USING (product_code)
  GROUP BY campaign_name, product_code, product_name
), r AS (
  SELECT *, DENSE_RANK() OVER (PARTITION BY campaign_name ORDER BY
percentage_change DESC) AS campaign_rank
  FROM t
)
SELECT * FROM r WHERE campaign_rank <= 3;
```

Output | Result #19 | Result #20

campaign_name	product_name	total_before	total_after	quantity_change	percentage_change	campaign_rank
Diwali	Atliq_waterproof_Immersion_Rod	2015	6950	4935	244.91	1
Diwali	Atliq_Curtains	2680	9214	6534	243.81	2
Diwali	Atliq_High_Glo_15W_LED_Bulb	3215	11053	7838	243.79	3
Sankranti	Atliq_Suflower_Oil (1L)	16257	61185	44928	276.36	1
Sankranti	Atliq_waterproof_Immersion_Rod	4453	16735	12282	275.81	2
Sankranti	Atliq_High_Glo_15W_LED_Bulb	5030	18875	13845	275.25	3

OK, 6 records retrieved in 8.759ms

```
-- Q20
WITH t AS (
  SELECT product_name, category,
         COUNT(*)                        AS event_count,
         SUM(`quantity_sold(before_promo)`) AS total_before,
         SUM(`quantity_sold(after_promo)`)  AS total_after,
         SUM(`quantity_sold(after_promo)`) - SUM(`quantity_sold(before_promo)`) AS
quantity_change,
         ROUND((SUM(`quantity_sold(after_promo)`) - SUM(`quantity_sold(before_promo)`))
             / NULLIF(SUM(`quantity_sold(before_promo)`), 0) * 100, 2) AS
percentage_change,
         SUM(base_price * `quantity_sold(before_promo)`) AS revenue_before,
         SUM(base_price * `quantity_sold(after_promo)`)  AS revenue_after,
         SUM(base_price * `quantity_sold(after_promo)`)
           - SUM(base_price * `quantity_sold(before_promo)`) AS revenue_change,
         ROUND(AVG(base_price), 2) AS avg_base_price
  FROM fact_events f
  JOIN dim_products p USING (product_code)
  GROUP BY product_code, product_name, category
), r AS (
  SELECT *, DENSE_RANK() OVER (PARTITION BY category ORDER BY revenue_change
DESC) AS category_rank
  FROM t
)
SELECT * FROM r WHERE category_rank <= 2
ORDER BY category, category_rank;
```

Output | Result #20

product_name	category	event_count	total_before	total_after	quantity_change	percentage_change	revenue_before	revenue_after	revenue_change	avg_base_price	category_rank
Atliq_Home_Essential_8_Product_Combo	Combo1	100	22299	63180	40881	183.33	66897000	189540000	122643000	3000.00	1
Atliq_Farm_Chakki_Atta (1KG)	Grocery & Staples	100	32340	81290	48950	151.36	10851800	29100500	18248700	330.00	1
Atliq_Sonamasuri_Rice (10KG)	Grocery & Staples	100	37281	53235	15954	42.79	32061660	45782100	13720440	860.00	2
Atliq_waterproof_Immersion_Rod	Home Appliances	100	6468	23685	17217	266.19	6597360	24158700	17561340	1020.00	1
Atliq_High_Glo_15W_LED_Bulb	Home Appliances	100	8245	29928	21683	262.98	2885750	10474800	7589050	350.00	2
Atliq_Double_Bedsheet_set	Home Care	100	4203	15058	10855	258.27	5001570	17919020	12917450	1190.00	1
Atliq_Curtains	Home Care	100	4592	16317	11725	255.34	1377600	4895100	3517500	300.00	2
Atliq_Doodh_Kesar_Body_Lotion (200ML)	Personal Care	100	5257	7022	1765	33.57	998830	1334180	335350	190.00	1
Atliq_Lime_Cool_Bathing_Bar (125GM)	Personal Care	100	7718	10280	2562	33.20	478516	637360	158844	62.00	2

OK, 9 records retrieved in 10.949ms